Anna-Maria Halacheva

PhD Researcher, INSAIT Sofia | 3D Computer Vision, Vision-Language Models, Embodied AI
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RESEARCH PROFILE

My research lies at the intersection of 3D computer vision, vision-language models, and embodied AI. I study how intelligent systems can develop scene intelligence: understanding physical structure, grounding language in 3D environments, and reasoning about space for future embodied agents.

EDUCATION

PhD Candidate in Computer Science / 3D Computer Vision | *INSAIT, Sofia University, Bulgaria* Sep 2024 - Present
Supervised by Prof. Luc Van Gool and Prof. Danda Paudel.

M.Sc. Computer Science | *University of Stuttgart, Germany* Oct 2022 - Oct 2024
First-class with Distinction, grade 1.2/1.0.

B.Sc. Software Engineering | *University of Stuttgart, Germany* Oct 2019 - Oct 2022
First-class, grade 1.4/1.0.

RESEARCH AND INDUSTRY EXPERIENCE

Graduate Research Fellow, 3D Vision | *INSAIT, Sofia University, Bulgaria* Apr 2024 - Aug 2024
- Developed the Articulate3D dataset of articulated 3D indoor scenes.
- Built annotation tooling for 3D segmentation, articulation, and visualization.
- Managed annotator hiring, supervision, data post-processing, and method development for object articulation prediction.

AI Engineer, Master@IBM | *IBM Germany, Ehningen* Jan 2023 - Apr 2024
Created machine learning and generative AI proof-of-concepts for Client Engineering customers.

SELECTED PUBLICATIONS AND RESEARCH PROJECTS

GaussianVLM: Scene-Centric 3D Vision-Language Models Using Language-Aligned Gaussian Splats for Embodied Reasoning and Beyond project/pdf
RA-L 2025 | Anna-Maria Halacheva, Jan-Nico Zaech, Xi Wang, Danda Pani Paudel, Luc Van Gool

Articulate3D: Holistic Understanding of 3D Scenes as Universal Scene Description project/pdf
ICCV 2025 | Anna-Maria Halacheva*, Yang Miao*, Jan-Nico Zaech, Xi Wang, Danda Pani Paudel, Luc Van Gool

From Scan to Action: Leveraging Realistic Scans for Embodied Scene Understanding project/pdf
CVPRW OpenSUN3D 2025 | Anna-Maria Halacheva, Jan-Nico Zaech, Sombit Dey, Danda Pani Paudel, Luc Van Gool

Expanding Expressivity in Transformer Models with MobiusAttention project/pdf
arXiv 2024 | Anna-Maria Halacheva, Mojtaba Nayyeri, Steffen Staab

SERVICE AND REVIEWING

OpenSUN3D Workshop Organization | *ICCV 2025 / CVPR 2026 / ECCV 2026* Dec 2024 - Present link
Second organizer, challenge chair, and program chair roles for the OpenSUN3D workshop series on open-world 3D scene understanding.

Master Thesis Supervision | *INSAIT* 2025
Supervision of one master's thesis student.

EXPLORER@INSAIT Supervisor | *INSAIT* Oct 2024 - Present
Supervision and mentorship of two undergraduate student researchers.

Reviewer Service | *CVPR, ECCV, NeurIPS, IROS, ICRA, RA-L* 2025 - 2026

Reviewer for major computer vision, robotics, and machine learning venues; recognized as a Top 5% Reviewer at CVPR 2026.

FELLOWSHIPS, AWARDS, AND RECOGNITION

Google DeepMind PhD Fellowship | *Google DeepMind* 2024 - Present

Featured Researcher, EU State of the Union Address | *European Commission* 2025 link

Featured as an example of European AI research effort.

Forbes Bulgaria 30 Under 30 | *Technology & Innovation* 2024

IBM Female Talents Network and Mentoring Program | *IBM Germany* 2022 - 2023

DAAD Undergraduate Fellowship | *German Academic Exchange Service* 2019 - 2022

AEB SE Fellowship | *AEB SE, Germany* 2021 - 2022

Bulgarian National Diploma | *Highest national distinction for high school graduates in Bulgaria* 2019

SELECTED PUBLIC ENGAGEMENT

The Superhuman Podcast with Georgi Nenov | *Conversation on AI research and Bulgaria's role in AI* 2026 link

Bulgarian National Radio | *Discussion on Bulgaria's place in the AI research landscape* 2026 link

KAS Panel Discussion | *Women's Empowerment event, Konrad Adenauer Stiftung* 2025 link

bTV National Television | *Discussion on AI and INSAIT's efforts in advancing the field* 2025 link